

Vineeth Raghav Gururajan
Stuttgart, Germany
+49 155 65548480
vineeth.raghav.g@gmail.com

SEG Automotive Germany GmbH
Human Resources
Stuttgart-Weilimdorf

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Application: Working Student (m/f/d) Hardware Architecture for Automotive Electronics

Dear Sophia Neubauer,

Good hardware starts with a good architecture: clear interfaces, documented concepts and requirements that can be traced through the product's life. Your Hardware Architecture working student role, which combines architecture models with ISO 26262 functional safety on real customer projects, is exactly the step I want to take. I am an M.Sc. Electrical Engineering student at the University of Stuttgart, with coursework in power systems, power electronics and battery modeling.

The posting asks for a basic understanding of electronic hardware, circuit design and system-level engineering, and I have worked with all three. During a one-year internship at FocalSystech, I defined electrical and hardware–software interfaces between sensors, actuators and microcontrollers (CAN, I2C, SPI, UART) and supported their integration into IoT devices. I designed and validated circuits, evaluated and selected components against technical requirements, and created PCB layouts in Altium. Defining an interface so that independent parts work together reliably is, to me, the heart of hardware architecture.

In a university team project on battery cell modeling, I analysed how electrical, thermal, aging and manufacturing tolerances affect cell behaviour and built that analysis into a simulation model. This taught me to represent a system in a structured, checkable way, which I expect to carry over to architecture models.

I want to be clear about what I have not done yet. I have not worked with Enterprise Architect or Cameo Systems Modeler, I have used MATLAB but not Simulink, and I have not applied ISO 26262 in practice. I am motivated to learn these, and I see your working student role as the right place to learn functional safety alongside hardware, system and safety engineers. My English is fluent. My German is at A2 and I am preparing for the telc B1 exam, and I am glad your posting treats German as a plus.

I can work up to 20 hours per week alongside my studies, with a start date by agreement. I look forward to discussing how I can support your hardware architecture team.

Kind regards,

Vineeth Raghav Gururajan

Enclosures: CV, proof of enrolment, certificates